

Paper 2 - Set 12 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	F	Straight lines
2	D	Polynomials, factor and remainder theorems
3	C	Quadratic functions
4	B	Polynomials, factor and remainder theorems
5	A	Identifying errors in proofs
6	B	Integration
7	B	Mathematical proof
8	B	Angles, polygons, congruence and vectors
9	D	The logic of arguments
10	D	Chains of reasoning
11	F	Circles and circle theorems
12	E	Chains of reasoning
13	A	Mathematical proof
14	D	Chains of reasoning
15	E	Identifying errors in proofs
16	A	Mathematical proof
17	H	Binomial expansion
18	E	Differentiation
19	D	Integration
20	A	Identifying errors in proofs

Worked solutions

Question 1 LIN-02 · Straight lines · spec MM3.1

The points $A(1, 2)$, $B(5, 4)$ and $C(k, 8)$ are collinear. Find k .

Answer: F

$$\text{Gradient } AB = \frac{2}{4} = \frac{1}{2}. \text{ Gradient } AC = \frac{6}{k-1} = \frac{1}{2} \Rightarrow k-1 = 12 \Rightarrow k = 13.$$

Question 2 POL-15 · Polynomials, factor and remainder theorems · spec MM1.6

The polynomial $x^3 + ax^2 + bx + c$ has roots 2, -1 and 3. Find the value of $a + b + c$.

Answer: D

$$p(1) = 1 + a + b + c, \text{ and } p(1) = (1-2)(1+1)(1-3) = (-1)(2)(-2) = 4. \text{ So } a + b + c = 3. \text{ (Directly: } (x-2)(x+1)(x-3) = x^3 - 4x^2 + x + 6.)$$

Question 3 QUA-24 · Quadratic functions · spec MM1.3

Let $f(x) = x^2 - 2x - 8$. For how many integers n is $f(n) < 0$?

Answer: C

$$(x-4)(x+2) < 0 \text{ gives } -2 < x < 4; \text{ the integers are } -1, 0, 1, 2, 3; \text{ five of them.}$$

Question 4 X2-08 · Polynomials, factor and remainder theorems · spec MM1.6

$p(x)$ is a polynomial. P : ' $p(2) = 0$ '. Q : ' $(x-2)^2$ is a factor of $p(x)$ '. Which one of the following is true?

Answer: B

If $(x-2)^2$ divides p then certainly $p(2) = 0$, so P is necessary for Q . But $p(x) = x-2$ has $p(2) = 0$ without a repeated factor, so P is not sufficient.

Question 5 ERR-01 · Identifying errors in proofs · spec Err2

A student attempts to solve the equation $\sqrt{x+3} = x-3$ for real x .

I Squaring both sides: $x+3 = (x-3)^2 = x^2 - 6x + 9$.

II So $x^2 - 7x + 6 = 0$.

III So $(x-1)(x-6) = 0$.

IV Hence the solutions are $x = 1$ and $x = 6$.

Which of the following best describes the solution?

Answer: A

Squaring can introduce extraneous solutions, so line IV should check them: $x = 1$ gives $\sqrt{4} = 2 \neq -2$. Only $x = 6$ works. Lines I-III are valid deductions.

Question 6 X2-17 · Integration · spec MM7.5

The trapezium rule with n strips is used to estimate $\int_0^1 \sqrt{x} \, dx$. Which of the following is true for every n ?

Answer: B

$y = \sqrt{x}$ is concave (bends downwards), so every chord lies below the curve and the trapezia underestimate the area.

Question 7 PRF-14 · Mathematical proof · spec Prf1

To prove **'if n^2 is even then n is even'** by contradiction, which assumption should be made?

Answer: B

Assume the hypothesis holds and the conclusion fails, then derive a contradiction: n odd gives n^2 odd, contradicting n^2 even.

Question 8 X2-20 · Angles, polygons, congruence and vectors · spec M5.3

$ABCD$ is a quadrilateral. P : 'the diagonals of $ABCD$ bisect each other'. Q : ' $ABCD$ is a rectangle'. Which one of the following is true?

Answer: B

Every rectangle is a parallelogram, and a parallelogram's diagonals bisect each other: P is necessary. A non-rectangular parallelogram has bisecting diagonals without being a rectangle: not sufficient.

Question 9 LGC-03 · The logic of arguments · spec Arg4

Which of the following is the negation of the statement **'every student in the class scored more than 50'**?

Answer: D

The negation of 'for all x , $P(x)$ ' is 'there exists x such that not $P(x)$ '. 'Not more than 50' means '50 or less'.

Question 10 RSN-12 · Chains of reasoning · spec Prf5

How many positive integers less than 1000 are divisible by neither 2 nor 5?

Answer: D

Of 1 to 999: 499 multiples of 2, 199 of 5, 99 of 10. Divisible by 2 or 5: $499 + 199 - 99 = 599$. Neither: $999 - 599 = 400$.

Question 11 CIR-11 · Circles and circle theorems · spec MM3.2

Two circles have equations $x^2 + y^2 = 25$ and $(x - 8)^2 + y^2 = 9$. Which of the following statements is/are true?

- I The circles touch externally.
- II The circles intersect at exactly two points.
- III The point $(5, 0)$ lies on both circles.

Answer: F

Radii 5 and 3, centres 8 apart $= 5 + 3$, so the circles touch externally at one point (I true, II false). That point is $(5, 0)$: $25 = 25$ and $9 + 0 = 9$ (III true).

Question 12 RSN-04 · Chains of reasoning · spec Prf5

x and y are positive integers with $x^2 - y^2 = 45$. How many possible pairs (x, y) are there?

Answer: E

$(x - y)(x + y) = 45$ with $0 < x - y < x + y$: $(1, 45)$, $(3, 15)$, $(5, 9)$ give $(x, y) = (23, 22)$, $(9, 6)$, $(7, 2)$. Three pairs.

Question 13 PRF-11 · Mathematical proof · spec Prf3

A student notices that $1 + 2 = 3$, $4 + 5 + 6 = 7 + 8$, $9 + 10 + 11 + 12 = 13 + 14 + 15$ and conjectures that the pattern continues. Which of the following describes the n th line of the pattern?

Answer: A

Line 1 starts at 1, line 2 at 4, line 3 at 9: start n^2 . Left sides have 2, 3, 4 numbers $(n + 1)$, right sides 1, 2, 3 (n) .

Question 14 RSN-01 · Chains of reasoning · spec Prf5

A selection of n distinct integers is taken from $1, 2, \dots, 20$. What is the smallest n that guarantees two of the chosen integers differ by exactly 10?

Answer: D

Pair up $\{1, 11\}, \{2, 12\}, \dots, \{10, 20\}$: ten pairs. Choosing 11 numbers forces two from the same pair (pigeonhole); choosing one from each pair gives 10 numbers with no such difference.

Question 15 ERR-17 · Identifying errors in proofs · spec Err1

Claim: if $f'(a) = 0$ and $f''(a) = 0$ then f has a point of inflexion at $x = a$.

Which of the following shows the claim is false?

Answer: E

x^4 has $f'(0) = f''(0) = 0$ but a minimum at 0, not an inflexion. (x^3 does have an inflexion; x^2 has $f''(0) \neq 0$.)

Question 16 PRF-09 · Mathematical proof · spec Prf1

Here is a proof that there are infinitely many primes, with the steps out of order.

- I Then N leaves remainder 1 on division by each of $p_1, \dots, p_n$, so no p_i divides N .
- II Suppose there are only finitely many primes $p_1, p_2, \dots, p_n$, and let $N = p_1 p_2 \dots p_n + 1$.
- III So N has a prime factor that is not in the list, which is a contradiction.
- IV But $N > 1$, so N has at least one prime factor.

Which is the correct order?

Answer: A

Define N (II), show no listed prime divides it (I), note that N nevertheless has some prime factor (IV), and conclude that this factor is a new prime (III). The conclusion III cannot come before both I and IV have been established.

Question 17 BIN-09 · Binomial expansion · spec MM2.4

Which of the following statements is/are true?

- I $\binom{n}{r} = \binom{n}{n-r}$ for all integers $0 \leq r \leq n$
- II $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ for all integers $0 \leq r < n$
- III $\sum_{r=0}^n \binom{n}{r} = 2^n$ for all positive integers n

Answer: H

I is the symmetry of Pascal's triangle; II is Pascal's rule (each entry is the sum of the two above it); III is $(1 + 1)^n$ expanded. All true.

Question 18 DIF-11 · Differentiation · spec MM6.3

Which of the following statements about $f(x) = x^3 - 3x$ is/are true?

- I f is increasing for all $x > 1$.
- II f has a local maximum at $x = -1$.
- III $f(x) \geq -2$ for all real x .

Answer: E

$f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$: positive for $x > 1$ (I true); $f'' = 6x < 0$ at $x = -1$ (II true). III false: the local minimum $f(1) = -2$ is not a global minimum since $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ (e.g. $f(-3) = -18$).

Question 19 X2-03 · Integration · spec MM7.1

f is a function defined for all real x and $\int_0^4 f(x) dx = 0$. Which of the following statements must be true?

- I $f(x) = 0$ for some x with $0 \leq x \leq 4$.
- II The area between the curve and the x -axis from 0 to 4 is 0.
- III $\int_0^2 f(x) dx = -\int_2^4 f(x) dx$

Answer: D

III is just the rule for combining integrals over adjacent ranges. I fails for a function that jumps from -1 to 1 without passing through 0 (nothing says f is continuous). II fails: $f(x) = x - 2$ has integral 0 but area 4.

Question 20 ERR-06 · Identifying errors in proofs · spec Err1

Claim: the function $f(x) = x^3 - 3x^2 + 3x$ is increasing for all real x .

- I $f'(x) = 3x^2 - 6x + 3 = 3(x-1)^2$.
- II So $f'(x) \geq 0$ for all x , with equality only at $x = 1$.
- III Since $f'(x) > 0$ for all x , f is increasing.

Which of the following is true?

Answer: A

f is (strictly) increasing because $f'(x) \geq 0$ with equality only at an isolated point, so $f(a) < f(b)$ whenever $a < b$; but line III's assertion ' $f'(x) > 0$ for all x ' is false at $x = 1$. There is no turning point at $x = 1$ (the derivative does not change sign).